



Solubility of dioxygen in seven fluorinated liquids

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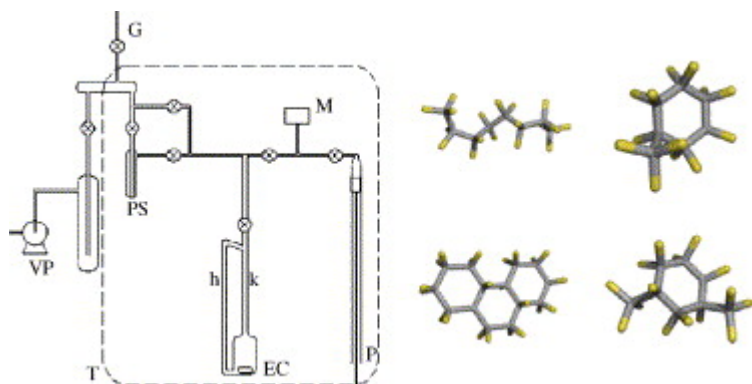
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Abstract

Experimental values for the solubility of oxygen in seven fluorinated organic liquids with potential for bio-medical applications—perfluorohexylethane, perfluoromethylcyclohexane, perfluorodimethylcyclohexane, perfluorophenanthrene, perfluorooctane, perfluorodecalin, and perfluorooctylbromide—are reported as a function of temperature and at pressures close to atmospheric. An apparatus based on a saturation method at constant pressure was used to measure solubility at temperatures from 288 to 316K. The precision of the results, expressed as Ostwald coefficients, is estimated as $\pm 1\%$. The solubility of oxygen is constant with temperature for perfluorooctylbromide, perfluorodimethylcyclohexane, perfluorohexylethane, perfluorodecalin and perfluorophenanthrene and decreases with temperature for perfluorooctane and perfluoromethylcyclohexane. Some of the experimental results are compared with existing literature values at 298K.



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Introduction

Preparations containing fully or partially fluorinated organic molecules attract attention from different fields of application ranging from chemical engineering to medicine. Recent applications include the use of fluorinated species in organic synthesis and in separation methods [1], because they are poorly miscible with both hydrocarbons and water [2], [3]; as agents to improve the solubility of chemicals in supercritical reaction or extraction media [4] and in biomedicine as gas carriers, as part of formulations for drug delivery or as ultrasound contrast agents [5].

The use of fluorinated substances in medical applications is largely determined by the combination of their high chemical stability with their potential to dissolve large quantities of gases. Chemical inertness is the precondition to fulfill the requirements of bio-compatibility which can only be attained in highly-fluorinated compounds when they are of high purity grade. Toxic by-products and impurities formed during manufacturing must be completely separated, in particular fluoroolefines formed, with hydrogen fluoride, due to the presence of residual hydrogen atoms in the structure. The capacity of fluorinated compounds to dissolve oxygen and carbon dioxide permit their use for the transport of gases in living organisms (artificial blood, liquid ventilation), and also in organ or tissue preservation during transportation. These processes require a correct calibration and control of the oxygen content, which is usually done by determining the partial pressure of gas. The link to the oxygen capacity—the most important parameter for preparations used as carrier systems for oxygen—requires exact data on gas solubility.

Literature data on oxygen solubility are still scattered and frequently available only at one temperature [5], [6], [7]. Some of the described structure-solubility relationships [8], [9], [10] enable a rough assessment of the performance of such preparations but these estimates are often not suitable for approving medical devices or drugs. The incomplete

characterization of the samples used and the imprecise description of the experimental techniques or prediction schemes employed make it difficult to compare published data, therefore limiting the use of such results.

The aim of the present work is to determine experimentally the solubility of oxygen in several fluorinated liquids of medical grade as a function of temperature. The range of temperatures covered is in relation with the biomedical applications of these solvents.

Further objectives of this study include the understanding of the dependence of macroscopic thermodynamic properties of solution on the underlying molecular interactions. A promising route to achieve this goal is the use of molecular simulation [11], [12]. Estimations have been considerably improved by the recent development of reliable molecular force-fields [13], [14], providing results as good as those of empirical methods but being predictive and rendering, in addition, microscopic structural details of the solutions [15].

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Results and discussion

Table 1 shows the experimental solubility of oxygen in perfluorinated liquids, between 288 and 318 K, expressed as Ostwald coefficients. The pressures indicated are values at equilibrium measured during experiments. Although the values differ slightly from 0.1 MPa this variation is considered to affect negligibly the values of oxygen solubility.

It is observed that, except for the case of perfluorooctane and perfluoromethylcyclohexane, in the other five solvents oxygen solubility does not vary ...

Materials

All the fluorinated liquid solvents used, perfluorohexylethane, $C_6F_{13}C_2H_5$, perfluoromethylcyclohexane, $C_6F_{11}CF_3$, perfluoro(1,3-dimethylcyclohexane), $C_6F_{10}(CF_3)_2$, perfluorophenanthrene, $C_{14}F_{24}$, perfluorooctane, C_8F_{18} , perfluorodecalin, $C_{10}F_{18}$ and

perfluorooctylbromide, C₈F₁₇Br, were obtained from Pharmpur GmbH, Germany. The samples used were purified following the same procedure in order to obtain an equivalent level of purity for each compound tested [24]. The different species fulfill the ...

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